

Smart Appliance Scheduler: Time-series NN based appliance scheduler for Cost-Optimal Usage

Ritvik Garimella
University of South Carolina
ritvikg@sc.edu

Abstract—The rapid growth of industrial automation and increasing global population has led to an increasing number of energy dependent devices, intensifying global electricity demand. Simultaneously, volatility in oil markets and uncertainty in electricity generation have contributed to rising and unpredictable energy costs. Moreover, the temporal mismatch between peak energy production and consumption often results in inefficiencies and transmission losses. To address these challenges, this paper proposes a time-series neural network based appliance scheduler for cost-optimal appliance scheduling in smart homes. The model analyses time-series data of electricity prices and household energy consumption along with individual appliance based energy limits with classification into three operational categories - base (will run regardless of systems like Wi-Fi routers, Security cameras, Refrigerators etc), negotiable (systems that can run later during time like washing machines, dish-washers, etc.), and non-negotiable (systems that user runs at their necessary convenience like Televisions, Water heaters, HVAC, etc.), based on their run-time flexibility. By recommending optimal operating windows for negotiable appliances, the proposed system aims to minimize consumer energy costs and promote balanced load distribution across the grid. Experimental results demonstrate the potential of intelligent scheduling as a practical step toward sustainable and cost-efficient energy management in smart grids with average savings that reduce the total electric cost by 28%.

Codes and Reports available at: https://github.com/Ritvik-G/USC_CSCE584/tree/main/MAS

I. INTRODUCTION

Residential electricity demand has increased substantially with the proliferation of connected devices and smart home devices, while the integration of renewable energy sources and dynamic pricing mechanisms has introduced significant short-term price volatility [1]. This volatility creates both challenges and opportunities for demand-side management: households face uncertain electricity costs, yet flexible appliance scheduling can exploit price fluctuations to reduce expenditure and alleviate peak grid stress.

Traditional residential scheduling approaches rely on static time-of-use tariffs, manually configured rules, or simple heuristics that cannot adapt to rapidly changing price signals or accommodate diverse appliance constraints [2]. Household appliances exhibit varying operational flexibility, some run continuously (e.g., refrigerators), others operate on user-initiated cycles (e.g., ovens, televisions), and a subset permit flexible scheduling within user-defined windows (e.g., washing machines, dishwashers). Effective scheduling systems must account for these distinctions to ensure feasibility and user acceptance.

Recent advances in time-series neural networks have improved the accuracy of short-term electricity price and load forecasting. Feed-forward and recurrent architectures capture nonlinear temporal dependencies more effectively than conventional statistical models [3], [4]. When integrated with optimization-based scheduling logic, these predictions enable intelligent load shifting toward low-price intervals, reducing consumer costs while supporting grid stability.

We propose a multi-agent framework that combines neural price and power forecasting with appliance-level profiling to generate cost-optimal schedules for flexible household loads. Appliances are categorized into essential (continuous operation), necessary (user-initiated), and expendable (schedulable) classes to enforce realistic operational constraints. The system employs lightweight neural models to forecast 24-hour price and baseline power profiles, then applies linear programming or greedy heuristics to determine optimal start times for expendable appliances subject to power limits.

The primary contributions are threefold. First, we present an integrated forecasting and scheduling architecture that unifies price prediction, load profiling, and constraint-based optimization. Second, we demonstrate through experiments on public datasets that the proposed approach achieves substantial cost reductions (average 28% over baseline schedules) while respecting capacity constraints. Third, we provide comparative analysis of exact (LP-based) and heuristic (greedy RL) scheduling methods, quantifying their respective trade-offs in solution quality and computational efficiency.

II. PROBLEM DESCRIPTION

Rising residential electricity demand, driven by automation and device proliferation, coupled with volatile energy markets and renewable integration, necessitates intelligent demand-side management strategies. The temporal misalignment between peak generation and consumer usage leads to inefficiencies, transmission losses, and increased cost during high-demand periods [5]. Appliance scheduling systems help mitigate these issues by shifting flexible loads toward lower-cost intervals, reducing household expenditure while easing grid congestion.

This work develops a neural network-based framework to forecast 24-hour electricity prices and household baseline load, followed by an optimization module that schedules flexible appliances using these predictions. Appliances are categorized according to scheduling flexibility:

- **Essential** ($\mathcal{A}_E$): operate continuously or on fixed schedules. In our case: *Fridge, Heater, Air Conditioning, Lights*.
- **Necessary** ($\mathcal{A}_N$): user-driven devices that cannot be autonomously shifted due to device constraints. In our case: *Oven, Microwave, TV, Computer*.
- **Expendable** ($\mathcal{A}_X$): appliances with flexible start times suitable for optimization. In our case: *Washing Machines, Dishwasher*.

The objective is to minimize the total operating cost of expendable appliances over a 24-hour horizon. Given forecasted hourly prices $\{\hat{p}_t\}_{t=1}^{24}$, baseline load $\{P_t\}_{t=1}^{24}$, appliance energy requirements $\{E(a)\}_{a \in \mathcal{A}_X}$, estimated durations $\{\hat{\tau}(a)\}_{a \in \mathcal{A}_X}$, and a household power limit $P_{\max}$, the goal is to assign an optimal start hour $\tau^*(a)$ for each flexible appliance:

$$\min_{\{\tau^*(a)\}} \sum_{a \in \mathcal{A}_X} E(a) \hat{p}_{\tau^*(a)}. \quad (1)$$

Feasibility is enforced by requiring that total consumption at each hour does not exceed the power limit. This formulation jointly enforces cost minimization and physical capacity limits, enabling feasible and economically efficient household scheduling.

III. RELATED WORK

Research on smart home energy management has focused on demand response (DR), appliance classification, forecasting, and optimization-based scheduling. Shewale *et al.* [1] provide a comprehensive survey of residential DR programs and optimization techniques, emphasizing the importance of flexible appliance categorization and intelligent scheduling for reducing peak demand and energy cost.

Several works employ advanced optimization frameworks for appliance scheduling. Maurya and Nanda [2] propose a fuzzy adaptive multi-objective scheduling method based on a dynamic SPEA2 algorithm, demonstrating improved cost-comfort trade-offs under time-varying prices. Shaban *et al.* [5] optimize appliance operation under inclining block rate tariffs, showing how tariff structure significantly influences household scheduling decisions.

Other studies investigate appliance classification and usage prediction as foundations for informed scheduling. Aslan and Zurel [6] present a hybrid time-series feature-based classification model for identifying appliance behavior patterns, while Basu *et al.* [7] demonstrate early time-series classification for predicting appliance usage trends.

Learning-driven approaches integrate forecasting with optimal control. Shafqat *et al.* [4] propose a comprehensive predictive-learning framework that combines user behavior modeling, appliance classification, and optimal scheduling. Yu *et al.* [3] further improve electricity usage prediction using a hybrid evolutionary optimization and Xception neural network architecture.

Collectively, these works show the relevance of combining appliance classification, forecasting, and optimization for cost-effective residential energy management. Building on this

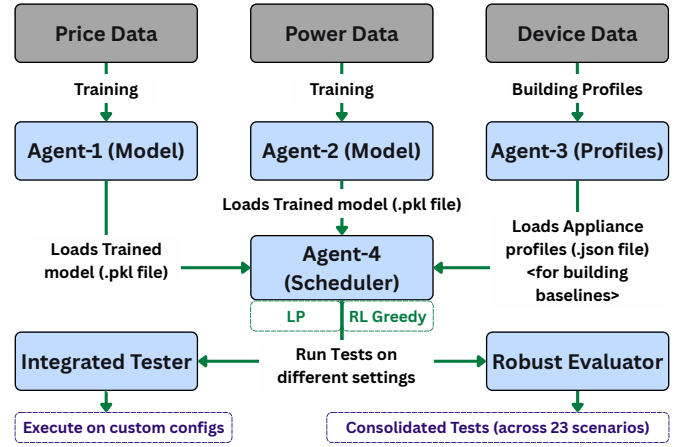


Fig. 1. Multi-Agent Architecture aimed at providing appropriate scheduling for expendable devices and suggestions for necessary devices. Agent 1 performs *Price Forecasting*, Agent 2 performs *Power Forecasting*, Agent 3 performs *Appliance Profiling* and Agent 4 is *Scheduling* the expendable appliances with time-slots within the selected day of the year with least power demand and cheapest cost during the day using two approaches (LP and Greedy RL). The integrated tester is available for custom value tests whereas robust evaluator runs across a pre-defined set of 23 testing scenarios to calculate effective price savings.

foundation, our work integrates time-series neural modeling with appliance flexibility categorization to provide cost-optimal scheduling recommendations for negotiable household loads.

IV. PROPOSED METHODOLOGY

A. System Overview

The proposed system consists of four interconnected agents:

- **Agent 1:** Electricity Price Forecasting
- **Agent 2:** Household Power Forecasting
- **Agent 3:** Appliance Profiling
- **Agent 4:** Scheduling Optimizer

Figure 1 illustrates the information flow between agents, where Agents 1-3 provide forecasts and profiles to Agent 4, which performs the final scheduling optimization.

B. Agent 1: Electricity Price Forecasting

1) *Problem Formulation:* Agent 1 predicts the electricity price $p_t \in \mathbb{R}^+$ (in \$/kWh) at hour t on day d , where $d \in [1, 365]$ and $t \in [1, 24]$.

2) *Feature Engineering:* To capture the cyclical nature of time, we employ sinusoidal transformations:

$$\mathbf{x}_{\text{hour}} = \begin{bmatrix} \sin\left(\frac{2\pi t}{24}\right) \\ \cos\left(\frac{2\pi t}{24}\right) \end{bmatrix} \quad \mathbf{x}_{\text{day}} = \begin{bmatrix} \sin\left(\frac{2\pi d}{365}\right) \\ \cos\left(\frac{2\pi d}{365}\right) \end{bmatrix} \quad (2)$$

The complete feature vector is:

$$\mathbf{x} = \left[\mathbf{x}_{\text{hour}}^T, \mathbf{x}_{\text{day}}^T, \frac{t}{24}, \frac{d}{365} \right]^T \in \mathbb{R}^6 \quad (3)$$

3) *Neural Network Architecture*: We employ a feed-forward neural network with the following architecture:

$$f(\mathbf{x}) = \mathbf{W}_4 \cdot \sigma(\mathbf{W}_3 \cdot \sigma(\mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3) + b_4 \quad (4)$$

where:

- Input layer: 6 neurons
- Hidden layer 1: 64 neurons
- Hidden layer 2: 32 neurons
- Hidden layer 3: 16 neurons
- Output layer: 1 neuron
- Activation function: $\sigma(z) = \max(0, z)$ (ReLU)

4) *Training Procedure*: The network is trained using the Adam optimizer to minimize the Mean Squared Error (MSE) between the predicted and true prices. The training process ran for 81 epochs before early stopping.

5) *Data Normalization*: Both features and targets are standardized using z-score normalization, where the mean and standard deviation are computed from the training set.

C. Agent 2: Household Power Forecasting

1) *Problem Formulation*: Agent 2 predicts household power consumption P_t (in kW) at hour t on day d .

2) *Enhanced Feature Engineering*: Building upon Agent 1's features, we add monthly cyclical encoding:

$$m = \left\lfloor \frac{d}{30.5} \right\rfloor, \quad \mathbf{x}_{\text{month}} = \begin{bmatrix} \sin\left(\frac{2\pi m}{12}\right) \\ \cos\left(\frac{2\pi m}{12}\right) \end{bmatrix} \quad (5)$$

The complete feature vector becomes:

$$\mathbf{x} = \left[\mathbf{x}_{\text{hour}}^T, \mathbf{x}_{\text{day}}^T, \mathbf{x}_{\text{month}}^T, \frac{t}{24}, \frac{d}{365} \right]^T \in \mathbb{R}^8 \quad (6)$$

3) *Data Aggregation*: Minute-level power consumption data is aggregated to hourly resolution:

$$P_{d,t} = \frac{1}{|\mathcal{T}_{d,t}|} \sum_{i \in \mathcal{T}_{d,t}} P_i \quad (7)$$

where $\mathcal{T}_{d,t}$ is the set of all minute-level readings within hour t of day d .

4) *Neural Network Architecture*: Similar to Agent 1, but adapted for 8 input features:

- Input layer: 8 neurons
- Hidden layers: $64 \rightarrow 32 \rightarrow 16$ neurons
- Output layer: 1 neuron
- Activation: ReLU
- Optimizer: Adam

The training process ran for 82 epochs before early stopping.

D. Agent 3: Appliance Profiling

1) *Appliance Categorization*: Appliances are grouped into three sets:

$$C(a) = \begin{cases} \text{ESSENTIAL}, & a \in \mathcal{A}_E \\ \text{NECESSARY}, & a \in \mathcal{A}_N \\ \text{EXPENDABLE}, & a \in \mathcal{A}_X \end{cases}$$

$$\mathcal{A}_E = \{\text{Fridge, Heater, AC, Lights}\}, \quad (8)$$

$$\mathcal{A}_N = \{\text{Oven, Microwave, TV, Computer}\}, \quad (9)$$

$$\mathcal{A}_X = \{\text{Washing Machine, Dishwasher}\}. \quad (10)$$

2) *Statistical Profile Construction*: For each appliance a , we compute summary statistics:

a) *Energy Statistics*:

$$E_{\text{avg}}(a) = \frac{1}{N_a} \sum_{i=1}^{N_a} E_i, \quad (11)$$

$$E_{\text{std}}(a) = \sqrt{\frac{1}{N_a - 1} \sum_{i=1}^{N_a} (E_i - E_{\text{avg}}(a))^2}. \quad (12)$$

b) *Hourly Usage Frequency*:

$$f_t(a) = \frac{|\{i : h_i = t, a_i = a\}|}{N_a},$$

c) *Peak Hours*:

$$\mathcal{H}_{\text{peak}}(a) = \text{argtop-}k_{t \in [1, 24]} f_t(a).$$

d) *Power and Duration Estimation*:

$$\hat{P}(a) = \frac{E_{\text{avg}}(a)}{\hat{\tau}(a)},$$

where the estimated duration is

$$\hat{\tau}(a) = \begin{cases} 2.0, & a \in \{\text{Washing Machine, Dishwasher}\} \\ 0.5, & a \in \{\text{Oven, Microwave}\} \\ 3.0, & a \in \{\text{TV, Computer}\} \\ 1.0, & \text{otherwise.} \end{cases}$$

E. Agent 4: Scheduling Optimizer

1) *Problem Formulation*: Given a set of appliances $\mathcal{A}$, price forecast $\{p_t\}$, baseline load $\{P_t\}$, appliance profiles $\{E(a), \hat{P}(a), \hat{\tau}(a)\}_{a \in \mathcal{A}}$, and a power limit P_{max} , the objective is to minimize total cost while respecting operational and power constraints. We evaluate two scheduling methods: a linear programming (LP) formulation and a greedy reinforcement learning (RL) heuristic.

2) *Scheduling Method-1: Linear Programming*:

a) *Decision Variables*:

$$x_{a,t} \in \{0, 1\}, \quad a \in \mathcal{A}, t \in [1, 24],$$

where $x_{a,t} = 1$ indicates that appliance a starts at hour t .

b) *Objective*:

$$\min \sum_{a \in \mathcal{A}} \sum_{t=1}^{24} x_{a,t} E(a) p_t$$

c) *Constraints.*:

- *single start*

$$\sum_{t=1}^{24} x_{a,t} = 1, \quad \forall a \in \mathcal{A}$$

- *Power limit.*:

$$P_t + \sum_{a \in \mathcal{A}} \sum_{\substack{s=1 \\ s \leq t < s + \lceil \hat{\tau}(a) \rceil}}^{24} x_{a,s} \hat{P}(a) \leq P_{\max}, \quad \forall t \in [1, 24]$$

This enforces that aggregate consumption at each hour stays within the capacity limit.

F. Reinforcement Learning Greedy Heuristic

a) *Algorithm.*: The RL-inspired heuristic schedules appliances in order of increasing predicted price. For each appliance, the algorithm selects the earliest feasible low-price hour, subject to the power limit and duration.

Greedy RL Scheduling

- 1: Sort hours by increasing price: $H_{\text{sorted}} = \text{argsort}(p_t)$.
- 2: Initialize schedule $S \leftarrow \emptyset$ and occupied hours $H_{\text{used}} \leftarrow \emptyset$.
- 3: **for** each $a \in \mathcal{A}$ **do**
- 4: $t_{\text{best}} \leftarrow \text{null}$, $c_{\text{best}} \leftarrow \infty$.
- 5: **for** each $t \in H_{\text{sorted}}$ **do**
- 6: **if** $t + \lceil \hat{\tau}(a) \rceil > 24$ or $t \in H_{\text{used}}$ **then**
- 7: **end if**
- 8: **if** $P_t + \hat{P}(a) > P_{\max}$ **then**
- 9: **end if**
- 10: $c \leftarrow E(a)p_t$.
- 11: **if** $c < c_{\text{best}}$ **then**
- 12: $c_{\text{best}} \leftarrow c$, $t_{\text{best}} \leftarrow t$.
- 13: **end if**
- 14: **end for**
- 15: **if** $t_{\text{best}} \neq \text{null}$ **then**
- 16: $S \leftarrow S \cup \{(a, t_{\text{best}})\}$.
- 17: Block hours $t_{\text{best}} \dots t_{\text{best}} + \lceil \hat{\tau}(a) \rceil - 1$ in H_{used} .
- 18: **end if**
- 19: **end for**
- 20: **return** S .

G. Baseline Schedule

A baseline is obtained by scheduling each appliance at its most frequent historical usage hour:

$$t_{\text{base}}(a) = \mathcal{H}_{\text{peak}}(a)[0], \quad C_{\text{base}} = \sum_{a \in \mathcal{A}} E(a) p_{t_{\text{base}}(a)} \quad (13)$$

H. Performance Metrics

We evaluate schedules using cost reduction (delta), relative savings (14.A), and LP–RL optimality gap (14.B):

$$\eta = \frac{\Delta C}{C_{\text{base}}} \times 100\%, \quad \gamma = \frac{C_{\text{RL}} - C_{\text{LP}}}{C_{\text{LP}}} \times 100\%. \quad (14)$$

V. EXPERIMENTAL SETUP

A. Datasets

We use three publicly available time-series datasets from Kaggle: (i) smart-home device-level consumption [8], (ii) California grid hourly prices [9], and (iii) household electricity records [10].

a) *Price forecasting.*: Hourly market prices over one year (8,760 hours) are used to train the price forecaster (Agent 1). Inputs are day-of-year and hour-of-day; the target is the hourly electricity price (converted from \$/MWh to \$/kWh).

b) *Power forecasting.*: Agent 2 uses smart home energy consumption data [8] resampled from minute-level to hourly resolution. This 6 month time-series data is used to create duck-curves for power consumption during a 24 hour period.

c) *Device consumption.*: Appliance-level logs from multiple homes [10] are used by Agent 3 to build appliance profiles (energy statistics, peak hours, and duration estimates) that would guide preference optimality of appliance's timings for baseline execution and simulations for tests.

B. Preprocessing and Splits

All datasets are aligned on timestamps and resampled to a uniform hourly frequency. We derive time-based features and aggregate category-wise loads where relevant. Continuous features and targets are standardized using z-score normalization computed on the training data. A chronological split is used throughout: 70% of the data for training, 15% for validation, and 15% for testing.

C. Scheduling Scenario

The scheduling agent (Agent 4) operates on a single target day (e.g., day 45, mid-February) using the price and baseline power forecasts from Agents 1 and 2 and the appliance profiles from Agent 3. We consider a set of schedulable appliances (Washing Machine, Dishwasher, Oven, Microwave, TV, Computer) and enforce a maximum power limit of $P_{\max} = 10$ kW. We compare three schedules: an exact LP-based optimizer, a greedy RL-inspired heuristic, and a baseline that runs each appliance at its most frequent historical usage hour.

D. Metrics

Forecasting performance (Agents 1 and 2) is evaluated using standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 . Scheduling performance (Agent 4) is measured by daily energy cost, percentage cost reduction relative to the baseline, and the optimality gap between the LP and RL schedules.

E. Implementation Details

All experiments are implemented in Python 3 using pandas and NumPy for data handling, scikit-learn for training the neural networks, and PuLP (CBC solver) for the LP-based scheduler. Random seeds are fixed for reproducibility.

In summary, this multi-agent setup does the following : agent 1 is doing price forecasting, agent 2 is doing power

TABLE I
FORECASTING PERFORMANCE OF AGENTS 1 (PRICE) AND 2 (POWER)

Task / Split	R^2	MAE	RMSE	MAPE (%)
Price-Train	0.8585	0.00378	0.00494	5.41
Price-Val	0.8596	0.00387	0.00504	5.49
Price-Test	0.8550	0.00384	0.00503	5.45
Power-Train	0.2772	0.667	0.863	—
Power-Val	0.1849	0.734	0.949	—
Power-Test	0.2964	0.648	0.849	—

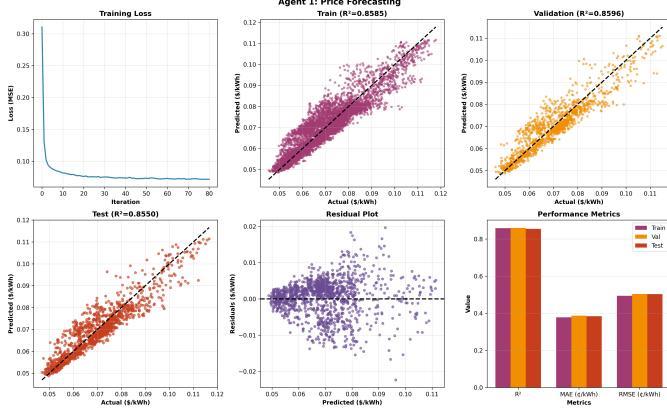


Fig. 2. Performance summary for **Agent 1: price forecasting** task. The figure presents: (i) the training loss curve illustrating convergence behavior; (ii) scatter plots of actual versus predicted prices for the training, validation, and test sets, each including the reference diagonal; (iii) a residual plot showing error distribution relative to predicted values; and (iv) a comparison of evaluation metrics (R^2 , MAE, RMSE) across all data partitions. Collectively, these results indicate consistent model performance and stable generalization across splits.

forecasting, agent 3 is doing appliance profiling and agent 4 is scheduling the expendable appliances with timeslots within the selected day of the year with least power demand and cheapest cost during the day.

VI. RESULTS

A. Forecasting Performance (Agents 1 and 2)

Table I summarizes the forecasting accuracy for electricity prices (Agent 1) and household power demand (Agent 2). Agent 1 achieves stable performance across splits ($R^2 \approx 0.86$, $MAE \approx \$0.0038/\text{kWh}$), providing sufficiently accurate short-term price signals for downstream scheduling (as shown in Figure 2). Agent 2 attains moderate accuracy ($R^2 = 0.18\text{--}0.30$) but captures demand peaks adequately for load-aware optimization (as shown in Figure 3).

B. Appliance Profiling and Baseline Construction (Agent 3)

No predefined household baselines are available; therefore, Agent 3 constructs them directly from appliance-level consumption data. It identifies ten appliances, of which two (Dishwasher, Washing Machine) are schedulable, and infers their operating durations, power ratings, temporal usage tendencies, and flexibility constraints. These data-derived profiles provide both the baseline load for cost evaluation and the flexibility

TABLE II
INTEGRATION TEST RESULTS FOR AGENT 4 (EXAMPLE: DAY 45)

Method	Cost (\$)	Savings (\$)	Savings (%)
Baseline	0.1543	—	—
LP	0.1116	0.0428	27.7
RL Greedy	0.1127	0.0416	27.0

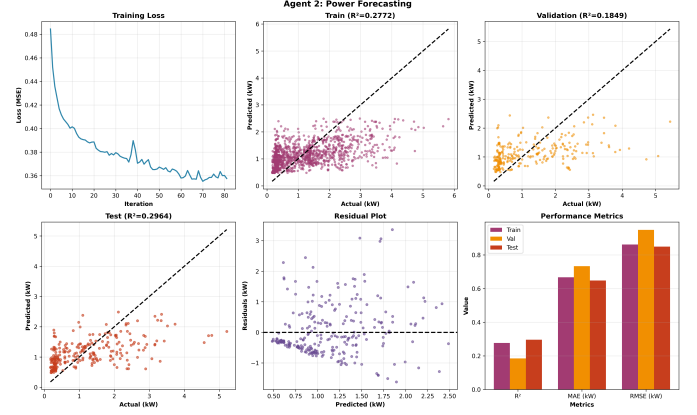


Fig. 3. Performance summary for **Agent 2: power forecasting**. The figure includes: (i) the training loss trajectory indicating gradual reduction in mean-squared error; (ii) scatter plots comparing actual and predicted power values for the training, validation, and test sets, each with a reference diagonal; (iii) a residual plot illustrating error distribution as a function of predicted output; and (iv) a bar chart reporting R^2 , MAE, and RMSE across the three data partitions. The results reflect limited predictive capability and higher variance relative to Agent 1, indicating greater modeling difficulty for this task.

constraints required by the scheduler (results shown in Figure 4).

C. Scheduling Performance — Integration Test (Agent 4)

Table II reports the results for a representative day (Day 45). Both Linear Programming (LP) and Reinforcement Learning (RL Greedy) shift appliance operation to lower-cost morning hours, reducing baseline cost by 27.7% and 27.0%, respectively. The LP-RL gap remains small (\$0.0011), demonstrating that the RL policy achieves near-optimal performance with lower computational complexity (results shown in Figure 5).

D. Robust Evaluation Across 23 Scenarios

The full pipeline (Agents 1–4) was evaluated across 23 heterogeneous scenarios that vary in price volatility, demand, appliance combinations, and temporal characteristics. Table ?? shows aggregated savings: LP achieves $24.97\% \pm 2.54\%$ mean savings, while RL achieves $24.21\% \pm 2.31\%$, recovering 99% of LP optimality on average. A paired t-test indicates a statistically significant but practically minor performance difference. Execution times remain below 70 ms for all cases.

E. Summary

Across all tests, the system (i) forecasts price and demand with sufficient accuracy, (ii) constructs realistic baseline loads from data, and (iii) achieves consistent household energy cost reductions of approximately 25%. LP provides guaranteed

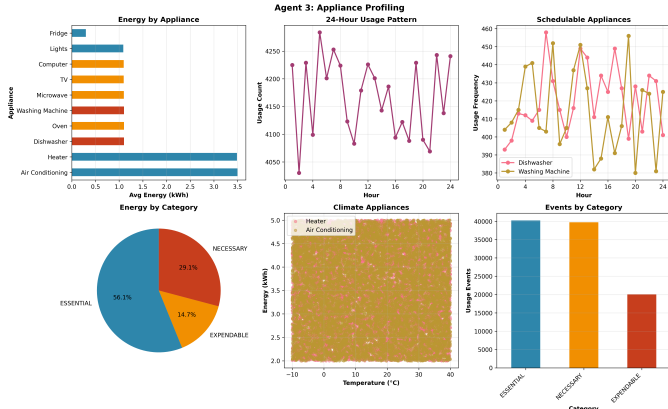


Fig. 4. Summary of **Agent-3: appliance-level behavioral profiling**. The figure shows: (i) average daily energy consumption across individual appliances; (ii) aggregate 24-hour usage patterns; (iii) hourly usage frequencies for schedulable appliances (dishwasher and washing machine); (iv) distribution of total energy consumption by appliance category (essential, necessary, expendable); (v) energy use of climate-related appliances as a function of ambient temperature; and (vi) total usage events by category. These results characterize device-specific consumption patterns and highlight opportunities for scheduling, load shifting, and demand-response optimization.

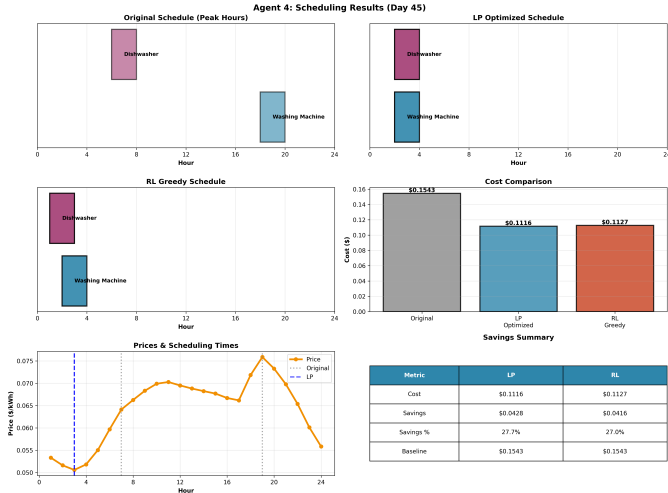


Fig. 5. Summary of **Agent 4 scheduling outcomes for example Day-45**. The figure presents: (i) the original appliance schedule during peak-price hours; (ii) the linear programming (LP) optimized schedule; (iii) the reinforcement learning (RL) greedy schedule; (iv) a cost comparison across the three schedules; (v) the daily price profile overlaid with original and LP-selected operation times; and (vi) a summary of cost savings achieved by LP and RL relative to the baseline. Results indicate that both optimization methods reduce operational cost, with LP yielding the highest savings for the evaluated day.

optimality, while RL offers a lightweight alternative with nearly identical performance.

VII. DISCUSSIONS AND CONCLUSIONS

The experimental results demonstrate that the proposed multi-agent framework achieves substantial residential electricity cost reductions of 25-28% through intelligent appliance scheduling. Agent 1's price forecasting exhibits robust performance ($R^2 \approx 0.86$), while Agent 2's power forecasting

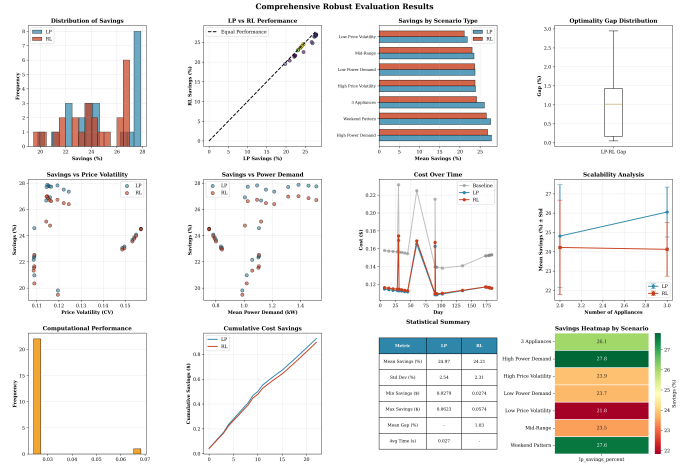


Fig. 6. **Robust evaluation results across 23 test scenarios**. The figure summarizes: (i) distributions of LP and RL savings; (ii) pairwise LP-RL performance comparison; (iii) mean savings by scenario type; (iv) optimality gap distribution; (v-vi) savings as a function of price volatility and power demand; (vii) cost trajectories over time; (viii) scalability with increasing appliance count; (ix) execution-time statistics; (x) cumulative cost savings; and (xi) aggregated statistical metrics. Overall, LP and RL exhibit consistent savings of approximately 25%, with RL achieving near-optimal performance and minimal computational overhead across diverse conditions.

TABLE III
ROBUST EVALUATION RESULTS ACROSS 23 DIVERSE SCENARIOS WRT BASELINE

Metric	LP	RL Greedy
Mean Savings (%)	24.97	24.21
Std Dev (%)	2.54	2.31
Min Savings (\$)	0.0279	0.0274
Max Savings (\$)	0.0623	0.0574
Optimality Gap (%)	0.00	1.03
Avg Execution Time (s)	0.027	-

($R^2 = 0.18-0.30$) proves adequate for load-aware scheduling. The minimal optimality gap between LP and RL methods ($\approx 1\%$) with execution times below 70ms validates both approaches for real-time applications. Consistency across 23 heterogeneous scenarios (mean 24-25% savings, std 2-3%) demonstrates robustness to demand patterns, price volatility, and appliance configurations. Key limitations include assumptions of perfect appliance controllability, optimization of only a subset of schedulable devices, and exclusive focus on cost minimization without modeling user comfort. Future extensions incorporating electric vehicles, battery storage, or multi-household coordination could amplify individual and system-wide benefits.

In conclusion, this work presents a modular four-agent architecture integrating time-series neural forecasting with constraint-aware optimization for practical smart home energy management. Future work should expand schedulable devices, incorporate multi-objective optimization, and validate through real-world deployments. As demand grows, intelligent management systems represent a critical pathway toward sustainable, cost-efficient smart grids that empower consumers while supporting grid stability.

REFERENCES

- [1] A. Shewale, A. Mokhade, N. Funde, and N. D. Bokde, "An overview of demand response in smart grid and optimization techniques for efficient residential appliance scheduling problem," *Energies*, vol. 13, no. 16, p. 4266, 2020.
- [2] V. K. Maurya and S. J. Nanda, "Time-varying multi-objective smart home appliances scheduling using fuzzy adaptive dynamic spea2 algorithm," *Engineering Applications of Artificial Intelligence*, vol. 121, p. 105944, 2023.
- [3] D. Yu, D. Wu, C. Liao, Z. Cao, and S. Pouramini, "Electricity usage prediction using developed human evolutionary optimization algorithm and xception neural network," *Scientific Reports*, vol. 15, no. 1, p. 24785, 2025.
- [4] W. Shafqat, K.-T. Lee, and D.-H. Kim, "A comprehensive predictive-learning framework for optimal scheduling and control of smart home appliances based on user and appliance classification," *Sensors*, vol. 23, no. 1, p. 127, 2022.
- [5] A. Shaban, M. Salhen, M. A. Shalaby, and T. F. Abdelmaguid, "Optimizing appliance scheduling for utility and budget in smart energy systems under inclining block rate tariffs," *Results in Engineering*, p. 106452, 2025.
- [6] M. Aslan and E. N. Zurel, "An efficient hybrid model for appliances classification based on time series features," *Energy and Buildings*, vol. 266, p. 112087, 2022.
- [7] K. Basu, V. Debusschere, and S. Bacha, "Appliance usage prediction using a time series based classification approach," in *IECON 2012-38th Annual Conference on IEEE Industrial Electronics Society*. IEEE, 2012, pp. 1217–1222.
- [8] M. Exwell, "Smart home energy consumption," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/mexwell/smart-home-energy-consumption>
- [9] T. Devastator, "California grid hourly prices," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/thedevastator/california-grid-hourly-prices>
- [10] Devastator, "240,000 household electricity consumption records," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/thedevastator/240000-household-electricity-consumption-records>